

LaTeX Cheat-Sheet — grounded in your CSE200 Online-1 documents

1. Document Skeleton

The frame every one of your files shares.

```
\documentclass[11pt]{article}
\usepackage[T1]{fontenc}
\usepackage[utf8]{inputenc}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath, amssymb, mathtools}
\usepackage{graphicx, xcolor, enumitem}

\title{Document Title}
\author{Name\ \small Affiliation}
\date{} % empty -> no date
\begin{document}
\maketitle
... content ...
\end{document}
```

% starts a comment. Blank line = new paragraph.

2. Sectioning

```
\section{The Puzzle in the Forest}
\subsection{A Firefly as a Clock}
\subsection*{Circuit Model} % * = unnumbered
\section*{Conclusion} % no number, no TOC
\tableofcontents \newpage % needs 2nd compile
```

Rename a heading (Q5): `\renewcommand{\contentsname}{Contents}`

3. Text & Fonts

```
\textbf{bold} \textit{italic} \emph{emphasis}
\underline{underlined} \texttt{monospace}
\textsc{Small Caps}
{\Large big} {\small small} {\huge huge}
\textcolor{blue}{coloured words}
super\textsuperscript{1} x\textsubscript{0}
\LaTeX \quad \quad ~ (non-break space)
--- (em dash) -- (en/range) ‘ ’ quotes
```

Size steps: `\tiny \scriptsize \footnotesize \small \normalsize \large \Large \LARGE`

4. Lists (enumitem)

```
\begin{itemize}
\item First point
\item[\textbf{Note}] custom label
\end{itemize}

\begin{enumerate}[label=\Roman*.] % I. II. III.
\item Drift
\end{enumerate}
```

% other labels: (*\Roman**) (*\alph**) (*\arabic**)

```
\begin{itemize}[label=$\diamond$] % custom bullet
\item ...
\end{itemize}
```

```
\begin{description}[style=nextline]
\item[Coupling] explanation on next line
\end{description}
```

Handy keys: `nosep`, `leftmargin=2em`, `label=$\oplus$`. Lists nest freely (see Q1/Q2).

5. Inline & Display Math

Inline: `$\theta_i(t)` `\in [0,2\pi]`, `$x_i$`, `$z^2$`.

Unnumbered display:
`\[ \Delta = \theta_2 - \theta_1 \]`

Numbered, with a label:
`\begin{equation}\label{eq:kuramoto}`
`T_i = \frac{2\pi}{\omega_i}`
`\end{equation}`

6. Multi-line Math

% aligned at @, each line numbered
`\begin{align}`
`c^2 &= a^2 - b^2 \\`
`e &= \frac{c}{a} \notag % \notag = no number`
`\end{align}`
`\begin{align*} ... \end{align*} % none numbered`

% one number for a multi-line derivation
`\begin{equation}`
`\begin{split}`
`\mathcal{L}\{1\} &= \int_0^\infty e^{-st} dt \\`
`&= \frac{1}{s}`
`\end{split}`
`\end{equation}`

% piecewise
`f(x)=\begin{cases}`
`x^2/\sigma^2 & \text{if } x \geq 0 \\`
`|x|/\sigma & \text{if } x < 0`
`\end{cases}`

7. Math Symbols You Use

Greek: `\theta \omega \sigma \mu \pi \alpha`
`\beta \gamma \Delta \Lambda \varpi \varepsilon`
Ops: `\sum_{j=1}^N \int_0^\infty \prod`
`\frac{a}{b} \dfrac{a}{b} \sqrt{x}`

```

\frac{\mathrm{d}\theta}{\mathrm{d}t}
Rel: \le \ge \approx \propto \equiv \in \pmod
\to \Rightarrow \implies \xrightarrow{L}
Deco: \bar x \hat x \vec x \dot x \bmod
\boxed{\Delta\omega} \le K
Cal: \mathcal{L} \mathbb{R} M_\odot \Gamma_\mu_\nu

```

`\,`, thin space, `\quad/\qquad` wide space, `\text{...}` for words inside math.

8. Tables

```

\begin{table}[ht]
\centering
\caption{Caption above or below}
\label{tab:x}
\begin{tabular}{|l|c|c|} % l/c/r + | rules
\hline
\multirow{2}{*}{Planet}
& \multicolumn{2}{c|}{Orbit} \\
\cline{2-3}
& $a$ & $e$ \\
\hline
Jupiter & 5.204 & 0.049 \\
\hline
\end{tabular}
\end{table}

```

booktabs (Q3): `\toprule` `\midrule` `\bottomrule` (no vertical bars).

Coloured rows (needs `\usepackage[table]{xcolor}`): `\rowcolor{green!10}`

Wrapping cell: column type `>\raggedright\arraybackslashp{4.3cm}`

9. Figures

```

\begin{figure}[ht]
\centering
\includegraphics[width=0.6\textwidth]{img.png}
\caption{...}\label{fig:x}
\end{figure}

```

Text wraps around (`wrapfig`, Q3/Q4):

```

\begin{wrapfigure}{r}{0.42\textwidth}
\centering
\includegraphics[width=0.40\textwidth]{f.jpg}
\caption{...}\label{fig:f}
\end{wrapfigure}

```

Side-by-side (`subcaption`, Q1/Q5):

```

\begin{subfigure}{0.3\textwidth}\centering
\includegraphics[width=\textwidth]{a}
\caption{Output A}
\end{subfigure}\hfill
\begin{subfigure}{0.3\textwidth} ... \end{subfigure}

```

`\hfill` spreads them; blank line starts a new row.

10. Cross-References

Give a `\label{...}` after the caption/equation, then refer to it anywhere:
`Table~\ref{tab:sine}` `fig.~\ref{fig:f}`
`eq.~\eqref{eq:kuramoto}` % (adds parentheses)
`Listing~\ref{lst:rc}`

`~` glues the word to the number so they never split across a line. *Always compile twice* to resolve numbers.

11. Colour (xcolor)

```

\definecolor{titleblue}{RGB}{40,80,170}
\textcolor{titleblue}{words}
\textcolor{green!55!black}{Positive} % mix
\rowcolor{red!10} % table row
red!70!black gray!15 green!45!black % blends

```

12. Code Listings (Q4)

```

\usepackage{listings}
\lstdefinestyle{pystyle}{
language=Python, basicstyle=\ttfamily\small,
keywordstyle=\color{blue!70!black}\bfseries,
numbers=left, frame=single, breaklines=true}

```

```

% pull code from an external file:
\lstinputlisting[style=pystyle,
caption={...}, label=lst:rc]{rc_response.py}

```

13. Theorems & Proof (Q5)

```

\usepackage{amsthm}
\newtheorem{proposition}{Proposition}[section]

\begin{proposition}[Weak field]\label{prop:x}
Statement of the result.
\end{proposition}
\begin{proof}
Argument. % gets the QED square
\end{proof}

```

14. Algorithms (Q3)

```

\usepackage{algorithm, algpseudocode}
\begin{algorithm}[ht]
\caption{Simulate the Kuramoto model}
\begin{algorithmic}[1]
Require Phases, $K$, step $h$, steps $M$
For{$m$ gets 1$} \textbf{to} $M$
\State $v_i$ gets $\omega_i + \frac{K}{N} q_i$
EndFor
\State \Return $\theta_1, \dots, \theta_N$
\end{algorithmic}
\end{algorithm}

```

15. Links, Footnotes, Citations

```
\usepackage{hyperref}
\hypersetup{colorlinks=true, linkcolor=blue,
            urlcolor=blue, citecolor=black}
\href{https://...}{clickable text}
A remark here.\footnote{The note text.}

Reference~\cite{sarfati2021}.
\begin{thebibliography}{9}
  \bibitem{sarfati2021}
```

```
Author. ‘‘Title’’. \emph{Journal} (2021).
\end{thebibliography}
```

16. Spacing & Boxes

```
\\[0.3em] break + extra vertical space
\vspace{1em} \bigskip \medskip \smallskip
\newpage \par \noindent \centering
\hfill (push apart) \quad \qquad (h-space)
\thispagestyle{empty} % no page number
```